Digital Twin Fidelity $\rightarrow$ Impact Analysis

What Level of Detail Is Sufficient?

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Outline

Motivation & Approach

Core Question

How does digital twin fidelity affect channel accuracy —
and what level of detail is **sufficient**?

Approach:

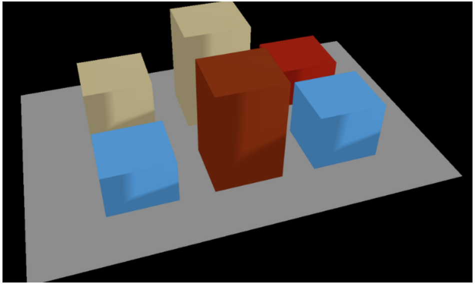
1. Build a high-fidelity **baseline** with Sionna RT.
2. Systematically **degrade** along 4 axes.
3. Measure **channel NMSE** at each level.
4. Identify the **transition point** where accuracy breaks.
5. Validate with **downstream ML tasks** (beam prediction, localization).

4 Degradation Axes

1. **Geometry**: building position/shape
2. **Material**: EM properties
3. **Ray Tracing**: depth, ray count
4. **Hardware**: antenna pattern, pol.

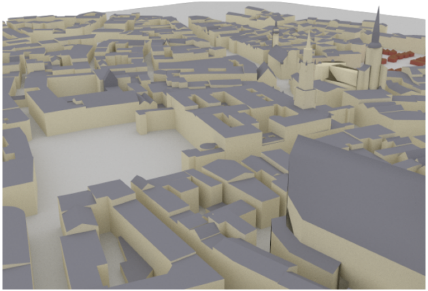
Evaluation Scenes

Simple Street Canyon



- 2601 UEs (51×51 grid, 0.8 m spacing), 1 TX.
- Uniform canyon geometry — clean multipath.
- **46/46 configs completed.**

Munich Urban Scene



- 2601 UEs, 332 buildings, diverse geometry.
- Realistic propagation with high scattering.
- **In progress** ($\sim$4 h/config $\times$ 46 $=$ 164 h).

46 Degradation Configs Across 4 Axes

1. Geometry (12 configs)

- Position noise: 0.1, 0.5, 1, 2, 3, 4, 5, 7, 10 m
- Height noise: 3 m
- Building removal: small (<5 m) or random 30%

2. Material (8 configs)

- Uniform override: concrete, glass, metal, wood, marble, brick
- Diffuse scattering on/off

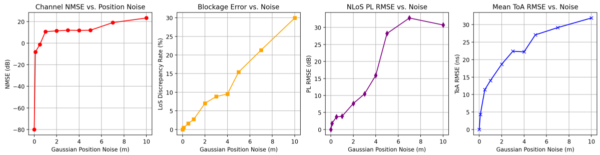
3. Ray Tracing (20 configs)

- Reflection depth: 0, 1, 2, . . . , 10
- Ray count: 1K, 5K, 10K, 20K, 50K, 100K, 200K, 500K
- Diffraction: 0 vs 1 event

4. Hardware (5 configs)

- Pattern: TR38.901 vs Dipole vs Isotropic
- Spacing: 0.5λ vs 0.4λ
- Polarization: V vs H

Geometry: Sub-meter Accuracy Is Critical

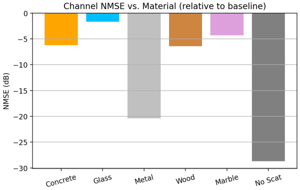


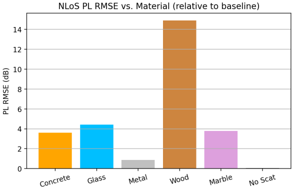
Position Noise	NMSE (dB)
0.1 m	-8.2
0.5 m	-1.3
1 m	$+10.6$
2 m	$+11.4$
5 m	$+12.0$
10 m	$+23.3$

Transition Point

$\leq$0.5 m: usable (-1.3 dB)
$\geq$1 m: **broken** ($> +10$ dB)

Material: Surprisingly Low Impact





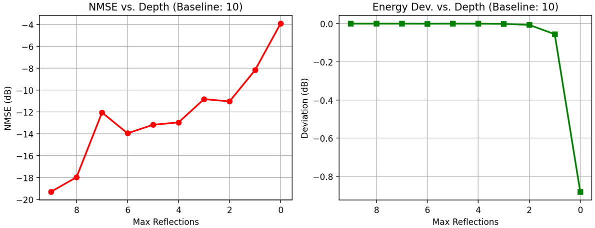
Material Config	NMSE (dB)
All concrete	-1.50
All glass	-0.73
All metal	-1.47
All wood	-1.69
All marble	-1.57
All brick	-1.46
No scattering	-1.47

Key Finding

Even replacing *all* materials → only $\approx -1.5\,\text{dB}$ NMSE.

Lowest priority in simple canyon.
Pending: Munich validation.

Ray Tracing — Reflection Depth: Diminishing Returns Above 5

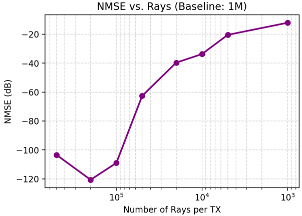


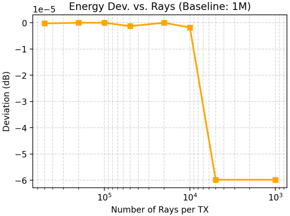
Max Depth	NMSE (dB)
0 (LoS only)	-3.9
1	-8.2
2	-11.0
3	-10.8
5	-13.2
7	-12.1
10 (baseline)	ref

Transition Point

Depth ≥ 3: adequate ($< -10\,$dB)
Depth ≥ 5: diminishing returns

Ray Tracing — Ray Count: 50K Is Enough



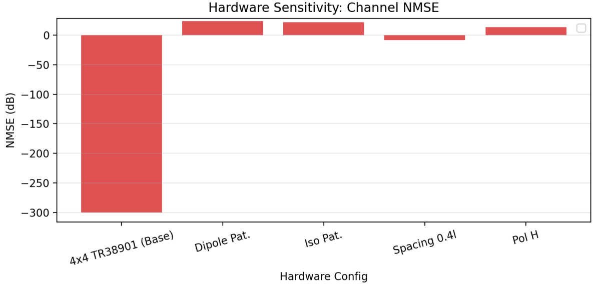


Ray Count	NMSE (dB)
1K	-12.1
5K	-20.5
50K	-35.3
200K	-40.6
1M (baseline)	ref

Transition Point

$\geq$50K rays $\approx$ baseline quality
20$\times$ speedup at negligible cost

Hardware: Antenna Pattern Is Make-or-Break



Hardware Config	NMSE (dB)
Isotropic	$+21.7$
Dipole	$+23.5$
Polarization H	$+13.5$
Spacing 0.4λ	-8.4
TR38.901 (baseline)	ref

Key Finding

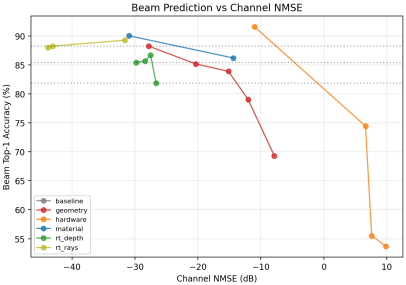
Wrong pattern $\rightarrow$ NMSE $> +20\,$dB.
As damaging as 10 m geometry noise.

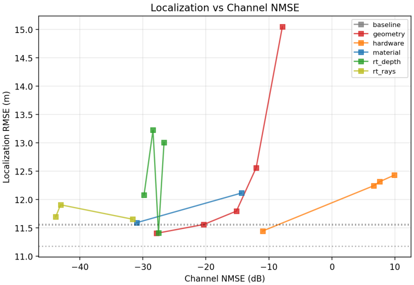
Realistic pattern modeling is mandatory.

Step 6: Do Channel Thresholds Hold for ML Tasks?

Experiment Design

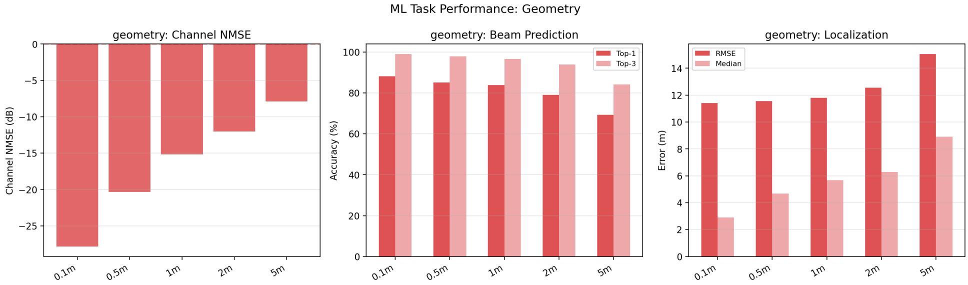
- **Train** beam prediction (position → beam) and localization (channel → position) on *degraded* DT data.
- **Test** on *baseline* data = simulated reality.
- Question: At what fidelity does ML task performance collapse?





Key insight: ML tasks are more tolerant than raw NMSE — geometry up to 2 m and RT depth=2 still yield >79% beam accuracy, while their channel NMSE is already degraded.

ML Results: Geometry — Gradual Degradation

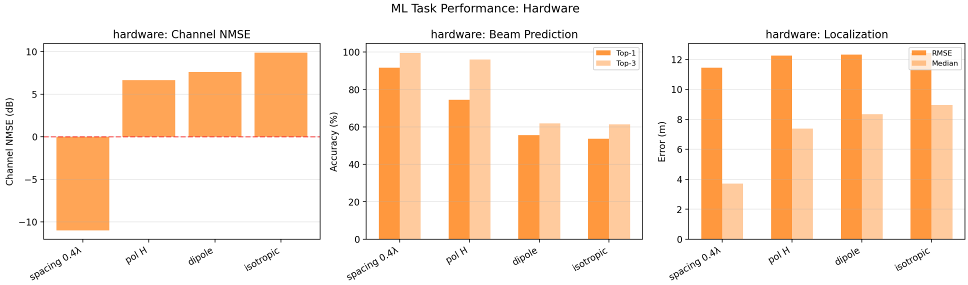


Noise	NMSE	Beam T1	Gain
0.1 m	-8.2	88.2%	94.7%
0.5 m	-1.3	85.2%	95.5%
1 m	$+10.6$	83.9%	91.3%
2 m	$+11.4$	79.0%	91.8%
5 m	$+12.0$	69.3%	82.1%

Finding

- Channel NMSE breaks at 1 m ($+10.6$ dB).
- But beam accuracy still 84% at 1 m.
- ML threshold is $\sim$2–5 m, **more tolerant** than channel NMSE.

ML Results: Hardware — Pattern Errors Are Fatal



Config	NMSE	Beam T1	Gain
Spacing	-8.4	91.6%	96.6%
Pol H	$+13.5$	74.4%	88.4%
Dipole	$+23.5$	55.5%	32.8%
Isotropic	$+21.7$	53.7%	29.6%

Finding

- Dipole/Isotropic: beam gain $\approx 30\%$.
- Model picks beams that capture only 1/3 of optimal power.
- **Pattern fidelity is non-negotiable** for both NMSE and ML.

ML Results: Material & Ray Tracing — Robust

Material

Config	NMSE	Beam T1	Gain
All concrete	-1.5	86.2%	96.0%
All metal	-1.5	90.0%	95.7%
Baseline	*ref*	*88.2%*	*97.4%*

Material changes have **no measurable ML impact**.

Uniform material assignment is acceptable.

Ray Tracing

Config	NMSE	Beam T1	Gain
Depth=2	-11.0	81.8%	89.7%
Depth=3	-10.8	85.7%	93.8%
Depth=5	-13.2	85.4%	93.1%
Depth=7	-12.1	86.7%	95.2%
5K rays	-20.5	88.2%	94.3%
50K rays	-35.3	88.0%	94.1%
Baseline	*ref*	*82–88%*	*93–97%*

Depth

≥ 3: beam accuracy within 3% of baseline.
Ray count: even 5K rays has **no ML impact**.

Answer: What Level of Detail Is Sufficient?

		Channel		ML Task	
Axis	**Threshold**	**NMSE**	**OK?**	**Beam**	**OK?**
Geometry ≤ 0.5 m	NMSE thr.	-1.3 dB	✓	85%	✓
Geometry ≤ 2 m	ML thr.	$+11.4$ dB	×	79%	✓
Material	Any	-1.5 dB	✓	86–90%	✓
RT Depth ≥ 3	Both	-10.8 dB	✓	86%	✓
RT Rays ≥ 5K	Both	-20.5 dB	✓	88%	✓
HW Pattern	TR38.901	ref	✓	86%	✓
HW Pattern	Dipole/Iso	$+22$ dB	×	54%	×
HW Pol.	Must match	$+13.5$ dB	×	74%	$\sim$
HW Spacing	$\pm 0.1\lambda$	-8.4 dB	✓	92%	✓

Key Takeaway

- **ML tasks are more tolerant than channel NMSE**: geometry 2 m still gives 79% beam accuracy despite $+11$ dB NMSE.
- **Exception — hardware**: antenna pattern errors break *both* channel and ML equally.
- **Biggest speedup opportunity**: ray count can be reduced $200\times$ (1M $\rightarrow$ 5K) with zero ML impact.

Real-World Validation: Setup

DICHASUS Dataset (dcxx)

- Real indoor channel measurements at 3.438 GHz, 50 MHz bandwidth.
- 64 distributed antennas (2 arrays of 4×8) in a hallway.
- 1024 OFDM subcarriers, tachymeter ground-truth positions.
- Source: Univ. Stuttgart, used in NVlabs' diff-RT calibration.

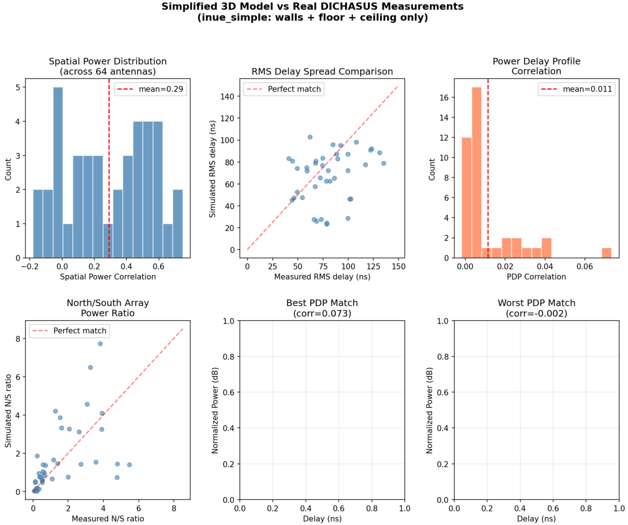
Simplified 3D Scene

- `inue_simple`: only 3 objects — walls, floor, ceiling.
- ITU material models (concrete, plasterboard, ceiling board).
- No furniture, pillars, doors, columns, or other scatterers.
- **Analogous to severe geometry degradation** in our framework.

Experiment

- Sionna 2.0 PathSolver: max_depth=5, 500K samples.
- 40 measurement positions, compare RT vs measured CSI.

Real-World Validation: Results



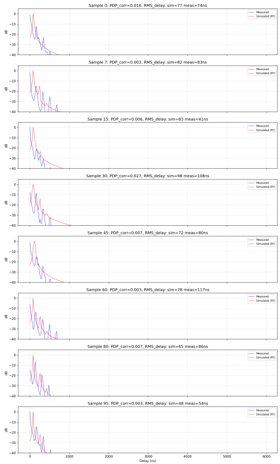
Real-World Validation: Key Findings

Metric	Value	Interpretation	Framework Prediction
Spatial power corr.	$r = 0.29$	Partial	Large-scale OK
RMS delay error	25 ns avg	Approximate	Envelope OK
PDP correlation	$r = 0.01$	None	Multipath fails
CFR correlation	$r = 0.05$	None	Channel fails
Scaled NMSE	$\approx 0\,\text{dB}$	Uncorrelated	Expected

Confirmation of Fidelity Framework

- **Geometry dominates**: simplified scene (walls only) captures large-scale power distribution but completely misses multipath structure — exactly as predicted by our synthetic experiments.
- **Consistent with Canyon results**: geometry degradation >2 m causes NMSE >+10 dB; the simplified INUE scene omits all interior objects (equivalent to ≫5 m error).
- **Next**: add detailed geometry from full 3D model to improve match, then apply degradation framework.

Power Delay Profile: Simulated vs Measured



Simplified scene captures general delay envelope but misses detailed multipath taps. Measured PDPs show richer scattering from objects absent in the 3-surface model.

Remaining Work

Completed

- ✓ Canyon: 46 configs, NMSE + ML analysis.
- ✓ Per-axis threshold table (channel + ML).
- ✓ Real-world validation with DICHASUS.

In Progress

- Munich 50×50: 46 configs (∼4 h/config, 164 h total).
- Detailed INUE scene from full .blend model.

Next Steps

- Build detailed 3D model → show improved NMSE.
- Apply fidelity degradation to INUE scene.
- **Key question**: Can calibrated RT on detailed scene achieve NMSE $< -10\,\mathrm{dB}$ vs real measurements?

Cross-Scene Validation

- Repeat analysis on Munich.
- Does material fidelity matter more in complex urban geometry?

Questions?